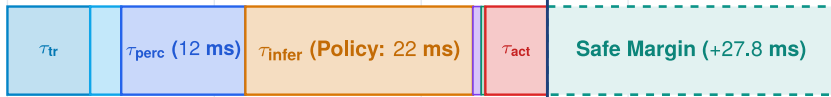


Pipeline Stages: τ_{tr} Transduce τ_{dma} DMA τ_{perc} Perception τ_{infer} Policy τ_{ipc} IPC τ_{enf} Enforce τ_{act} Actuator

Physical World Deadline: $\tau_{world} = 80.0$ ms

Nominal Execution ($P_{50} = 52.2$ ms)

52.2 ms (Safe)



Tail Latency Spike ($P_{99} = 126.0$ ms)

126.0 ms (Violation)



Critical Timing Breach: +46.0 ms over physical deadline!

